LABORATORY EXERCISE SUBMISSION

Lab 2: Custom Reusable Package with String & Math Functions (Exp 2)

Weightage: 5 marks
Core Concepts: Custom packages, identifier export rules (capitalization), multi-file modularity, Git submission.

1. Lab Tasks

1. Create a custom Go package containing two string functions (`Reverse`, `CountVowels`) and two math functions (`Factorial`, `Power`).
2. Exported function identifiers must begin with an uppercase letter.
3. Call all four functions from a separate `main.go` program demonstrating package import and reusability.
4. Push the package directory and main program to a GitHub repository and provide output.

2. Package Implementation (`utils/utils.go`)

Package: utils

```
package utils

import "strings"

// Reverse returns the reversed string (supports full rune unicode)
func Reverse(s string) string {
        runes := []rune(s)
        for i, j := 0, len(runes)-1; i < j; i, j = i+1, j-1 {
                runes[i], runes[j] = runes[j], runes[i]
        }
        return string(runes)
}

// CountVowels counts occurrences of standard vowels
func CountVowels(s string) int {
        vowels := "aeiouAEIOU"
        count := 0
        for _, c := range s {
                if strings.ContainsRune(vowels, c) {
                        count++
                }
        }
        return count
}

// Factorial computes n! for non-negative integers
func Factorial(n int) int64 {
        if n <= 1 { return 1 }
        var res int64 = 1
        for i := 2; i <= n; i++ { res *= int64(i) }
        return res
}

// Power computes base raised to integer exponent exp
func Power(base float64, exp int) float64 {
        res := 1.0
        for i := 0; i < exp; i++ { res *= base }
        return res
}
```

3. Main Driver Program (`main.go`)

File: main.go (Demonstrating Import & Reusability)

```
package main

import (
        "fmt"
        "lab2/utils"
)

func main() {
        sample := "Golang Development"
        fmt.Println("=== String Utility Functions ===")
        fmt.Printf("Original String : %s\n", sample)
        fmt.Printf("Reversed String : %s\n", utils.Reverse(sample))
        fmt.Printf("Vowel Count     : %d\n\n", utils.CountVowels(sample))

        fmt.Println("=== Math Utility Functions ===")
        fmt.Printf("Factorial(5)    : %d\n", utils.Factorial(5))
        fmt.Printf("Power(2.5, 3)   : %.4f\n", utils.Power(2.5, 3))
}
```

4. Execution Output & GitHub Submission

```
$ go run main.go
=== String Utility Functions ===
Original String : Golang Development
Reversed String : tnempoleveD gnaloG
Vowel Count     : 5

=== Math Utility Functions ===
Factorial(5)    : 120
Power(2.5, 3)   : 15.6250

$ git add go.mod main.go utils/ && git commit -m "Lab 2: Implement custom reusable package" && git push origin
main
To https://github.com/user/golang-labs.git
   f32b1a..c8724d  main -> main
```